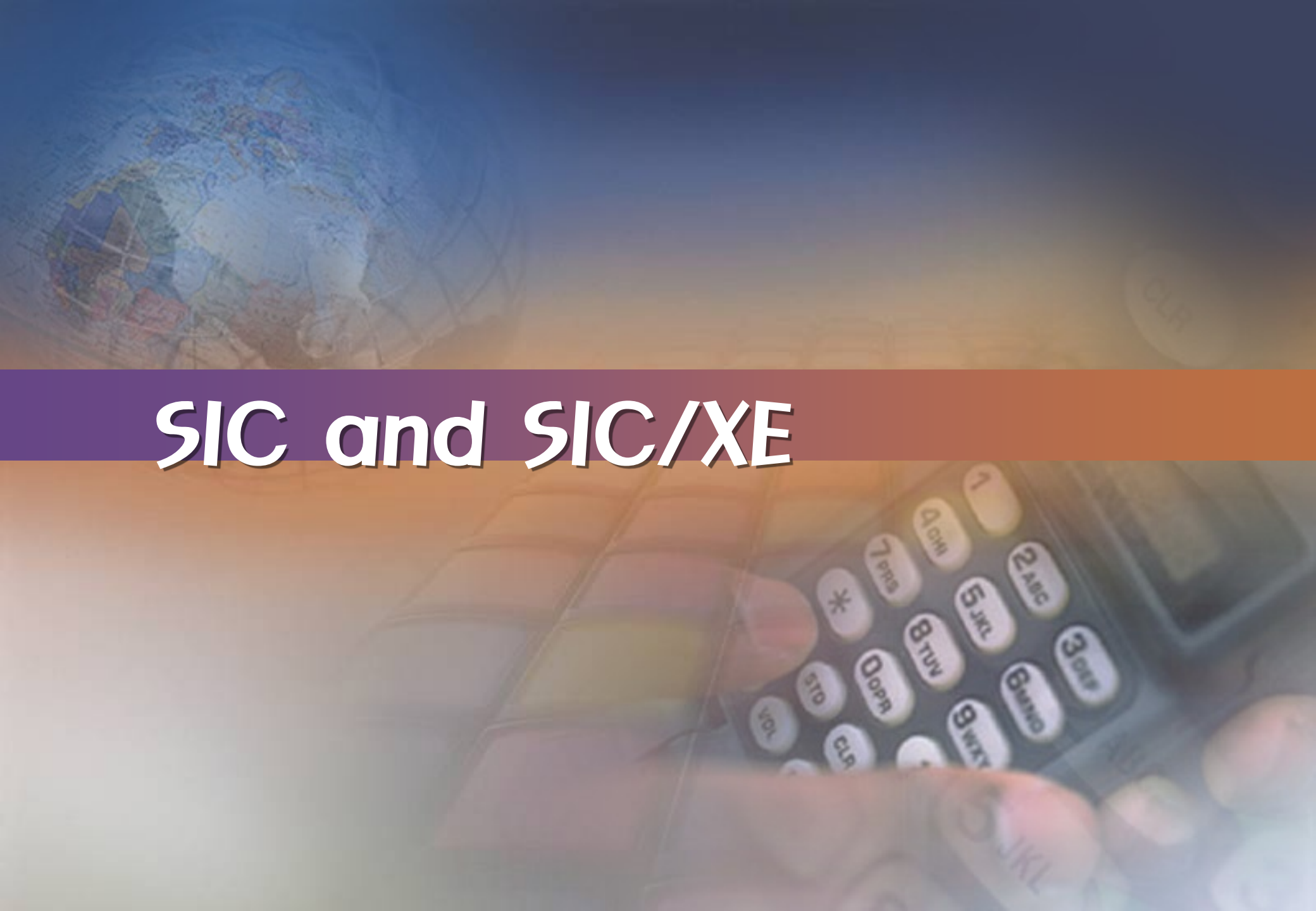


The background of the slide is a composite image. The upper left portion shows a stylized globe with a grid pattern, rendered in shades of blue and white. The lower right portion shows a close-up of a hand using a calculator, with the calculator's keypad and display visible. A horizontal band of purple and orange gradient separates the globe from the calculator.

SIC and SIC/XE

수업 목표

- SIC/XE 명령어의 사용법을 간단한 예제로 이해한다.
- SIC/XE 명령어의 주소 지정 방식과 사용방식에 대해 간단한 예제를 통해 확인한다.

Example

.	.	(B) = 006000
.	.	(PC) = 003000
.	.	(X) = 000090
3030	003600	
.	.	
.	.	
3600	103000	
.	.	
.	.	
6390	00C303	
.	.	
.	.	
C303	003030	
.	.	
.	.	
.	.	

Machine instruction						Value loaded into register A	
Hex		Binary					
							Target address
		op	n i x b p e	disp/address			
032600	000000	1 1 0 0 1 0	0110 0000 0000		3600	103000	
03C300	000000	1 1 1 1 0 0	0011 0000 0000		6390	00C303	
022030	000000	1 0 0 0 1 0	0000 0011 0000		3030	103000	
010030	000000	0 1 0 0 0 0	0000 0011 0000		30	000030	
003600	000000	0 0 0 0 1 1	0110 0000 0000		3600	103000	
0310C303	000000	1 1 0 0 0 1	0000 1100 0011 0000 0011	C303	003030		

Fig 1.1 Example of SIC/XE instructions and addressing modes

How to know the address

1. It is relative addressing or direct addressing

- check p and b flag bit
- use p or b with displacement or address for the TA

2. It is format 3 or format 4

- check e flag bit
- use displacement by 12bits or address by 20bits

3. It is index or not

- check x flag bit
- add index value with 2th result.

4. It is immediate addressing mode or indirect addressing mode or simple addressing mode

- check n and i flag bit

Example

	Machine instruction						Value loaded into register A
	Hex	Binary				Target address	
		op	n i x b p e	disp/address			
1)	032600	000000	1 1 0 0 1 0	0110 0000 0000		3600	103000
2)	03C300	000000	1 1 1 1 0 0	0011 0000 0000		6390	00C303
3)	022030	000000	1 0 0 0 1 0	0000 0011 0000		3030	103000
4)	010030	000000	0 1 0 0 0 0	0000 0011 0000		30	000030
5)	003600	000000	0 0 0 0 1 1	0110 0000 0000		3600	103000
6)	0310C303	000000	1 1 0 0 0 1	0000 1100	0011 0000 0011	C303	003030

- 1) pc-relative, format 3, no index, simple addressing mode -> $TA = pc + disp(12bits)$
- 2) base-relative, format 3, index, simple addressing mode -> $TA = pc + index + disp(12bits)$
- 3) pc-relative, format 3, no index, indirect addressing mode -> $TA = \text{value of the position of pc} + disp(12bits)$
- 4) format 3, no index, direct and immediate addressing mode -> $TA = address(12bit)$
- 5) format 3(If n and i = 0, then b, e and e flag bits are used as address in SIC by format 3) no index, direct addressing mode -> $TA = address(15bit)$
- 6) format 4, no index, direct and simple addressing mode -> $TA = address(20bits)$

SIC/XE instruction Set 1

Appendix A

SIC/XE Instruction Set and Addressing Modes

각각의 명령은
명령어, 형식, 16진코드,
의미로 구성된다.

Instruction Set

In the following descriptions, uppercase letters refer to specific registers. The notation m indicates a memory address, n indicates an integer between 1 and 16, and $r1$ and $r2$ represent register identifiers. Parentheses are used to denote the contents of a register or memory location. Thus $A \leftarrow (m..m+2)$ specifies that the contents of the memory locations m through $m+2$ are loaded into register A; $m..m+2 \leftarrow (A)$ specifies that the contents of register A are stored in the word that begins at address m .

The letters in the Notes column have the following meanings:

- P Privileged instruction
- X Instruction available only on XE version
- F Floating-point instruction
- C Condition code CC set to indicate result of operation (<, =, or >)

The Format column indicates which SIC/XE instruction format is to be used in assembling each instruction; 3/4 means that either Format 3 or Format 4 can be used. All instructions for the standard version of SIC are assembled using the format described in Section 1.3.1 (which is compatible with Format 3). Instruction subfields that are not required, such as the address field for an RSUB instruction, are set to zero.

Mnemonic	Format	Opcode	Effect	Notes
ADD m	3/4	18	$A \leftarrow (A) + (m..m+2)$	
ADDF m	3/4	58	$F \leftarrow (F) + (m..m+5)$	X F
ADDR r1,r2	2	90	$r2 \leftarrow (r2) + (r1)$	X
AND m	3/4	40	$A \leftarrow (A) \& (m..m+2)$	
CLEAR r1	2	B4	$r1 \leftarrow 0$	X
COMP m	3/4	28	$(A) : (m..m+2)$	C
COMPF m	3/4	88	$(F) : (m..m+5)$	X F C
COMPR r1,r2	2	A0	$(r1) : (r2)$	X C
DIV m	3/4	24	$A \leftarrow (A) / (m..m+2)$	
DIVF m	3/4	64	$F \leftarrow (F) / (m..m+5)$	X F
DIVR r1,r2	2	9C	$r2 \leftarrow (r2) / (r1)$	X
FIX	1	C4	$A \leftarrow (F)$ [convert to integer]	X F
FLOAT	1	C0	$F \leftarrow (A)$ [convert to floating]	X F
HIO	1	F4	Halt I/O channel number (A)	P X
J m	3/4	3C	$PC \leftarrow m$	
JEQ m	3/4	30	$PC \leftarrow m$ if CC set to =	
JGT m	3/4	34	$PC \leftarrow m$ if CC set to >	
JLT m	3/4	38	$PC \leftarrow m$ if CC set to <	
JSUB m	3/4	48	$L \leftarrow (PC); PC \leftarrow m$	
LDA m	3/4	00	$A \leftarrow (m..m+2)$	
LDB m	3/4	68	$B \leftarrow (m..m+2)$	X
LDCH m	3/4	50	A [rightmost byte] $\leftarrow (m)$	
LDF m	3/4	70	$F \leftarrow (m..m+5)$	X F
LDL m	3/4	08	$L \leftarrow (m..m+2)$	
LDS m	3/4	6C	$S \leftarrow (m..m+2)$	X
LDT m	3/4	74	$T \leftarrow (m..m+2)$	X
LDX m	3/4	04	$X \leftarrow (m..m+2)$	
LPS m	3/4	D0	Load processor status from information beginning at address m (see Section 6.2.1)	P X
MUL m	3/4	20	$A \leftarrow (A) * (m..m+2)$	

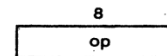
SIC/XE instruction Set 2

Mnemonic	Format	Opcode	Effect	Notes
MULF m	3/4	60	$F \leftarrow (F) * (m..m+5)$	X F
MULR r1, r2	2	98	$r2 \leftarrow (r2) * (r1)$	X
NORM	1	C8	$F \leftarrow (F)$ [normalized]	X F
OR m	3/4	44	$A \leftarrow (A) \mid (m..m+2)$	
RD m	3/4	D8	A [rightmost byte] $\leftarrow$ data from device specified by (m)	P
RMO r1, r2	2	AC	$r2 \leftarrow (r1)$	X
RSUB	3/4	4C	$PC \leftarrow (L)$	
SHIFTL r1, n	2	A4	$r1 \leftarrow (r1)$; left circular shift n bits. {In assembled instruction, $r2 = n-1$ }	X
SHIFTR r1, n	2	A8	$r1 \leftarrow (r1)$; right shift n bits, with vacated bit positions set equal to leftmost bit of (r1). {In assembled instruction, $r2 = n-1$ }	X
SIO	1	F0	Start I/O channel number (A); P X address of channel program is given by (S)	
SSK m	3/4	EC	Protection key for address m $\leftarrow (A)$ (see Section 6.2.4)	P X
STA m	3/4	0C	$m..m+2 \leftarrow (A)$	
STB m	3/4	78	$m..m+2 \leftarrow (B)$	X
STCH m	3/4	54	$m \leftarrow (A)$ [rightmost byte]	
STF m	3/4	80	$m..m+5 \leftarrow (F)$	X F
STI m	3/4	D4	Interval timer value $\leftarrow (m..m+2)$ (see Section 6.2.1)	P X
STL m	3/4	14	$m..m+2 \leftarrow (L)$	
STS m	3/4	7C	$m..m+2 \leftarrow (S)$	X
STSW m	3/4	E8	$m..m+2 \leftarrow (SW)$	P
STT m	3/4	84	$m..m+2 \leftarrow (T)$	X
STX m	3/4	10	$m..m+2 \leftarrow (X)$	
SUB m	3/4	1C	$A \leftarrow (A) - (m..m+2)$	
SUBF m	3/4	5C	$F \leftarrow (F) - (m..m+5)$	X F

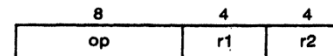
Mnemonic	Format	Opcode	Effect	Notes
SUBR r1, r2	2	94	$r2 \leftarrow (r2) - (r1)$	X
SVC n	2	B0	Generate SVC interrupt. {In assembled instruction, $r1 = n$ }	X
TD m	3/4	E0	Test device specified by (m)	P C
TIO	1	F8	Test I/O channel number (A)	P X C
TIX m	3/4	2C	$X \leftarrow (X) + 1$; (X): (m..m+2)	C
TIXR r1	2	B8	$X \leftarrow (X) + 1$; (X): (r1)	X C
WD m	3/4	DC	Device specified by (m) $\leftarrow (A)$ [rightmost byte]	P

Instruction Formats

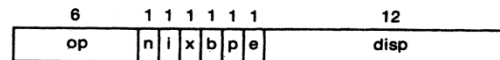
Format 1 (1 byte):



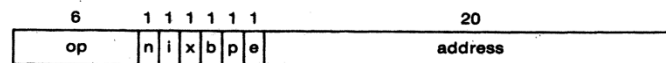
Format 2 (2 bytes):



Format 3 (3 bytes):



Format 4 (4 bytes):



Addressing Modes

The following addressing modes apply to Format 3 and 4 instructions. Combinations of addressing bits not included in this table are treated as errors by the machine. In the description of assembler language notation, *c* indicates a constant between 0 and 4095 (or a memory address known to be in this range); *m* indicates a memory address or a constant value larger than 4095. Further information can be found in Section 1.3.2.

SIC/XE 명령은 단일 모드를 사용하는 SIC 명령어와 달리 총4개의 명령어 형태를 가진다.

ASCII Character Codes

SIC/XE 명령의 플래그 비트가 주소 지정 방식을 결정한다.

Appendix B

ASCII Character Codes

The letters in the Notes column have the following meanings:

- 4 Format 4 instruction
- D Direct-addressing instruction
- A Assembler selects either program-counter relative or base-relative mode
- S Compatible with instruction format for standard SIC machine. Operand value can be between 0 and 32,767 (see Section 1.3.2 for details).

Addressing type	Flag bits n l x b p e	Assembler language notation	Calculation of target address TA	Operand	Notes
Simple	110000	op c	disp	(TA)	D
	110001	+op m	addr	(TA)	4 D
	110010	op m	(PC) + disp	(TA)	A
	110100	op m	(B) + disp	(TA)	A
	111000	op c,X	disp + (X)	(TA)	D
	111001	+op m,X	addr + (X)	(TA)	4 D
	111010	op m,X	(PC) + disp + (X)	(TA)	A
	111100	op m,X	(B) + disp + (X)	(TA)	A
	000---	op m	b/p/e/disp	(TA)	D S
	001---	op m,X	b/p/e/disp + (X)	(TA)	D S
Indirect	100000	op @c	disp	((TA))	D
	100001	+op @m	addr	((TA))	4 D
	100010	op @m	(PC) + disp	((TA))	A
	100100	op @m	(B) + disp	((TA))	A
Immediate	010000	op #c	disp	TA	D
	010001	+op #m	addr	TA	4 D
	010010	op #m	(PC) + disp	TA	A
	010100	op #m	(B) + disp	TA	A

Hex code	ASCII character	Hex Code	ASCII character	Hex code	ASCII character	Hex code	ASCII character
00	NUL	20	SP	40	@	60	`
01	SOH	21	!	41	A	61	a
02	STX	22	"	42	B	62	b
03	ETX	23	#	43	C	63	c
04	EOT	24	\$	44	D	64	d
05	ENQ	25	%	45	E	65	e
06	ACK	26	&	46	F	66	f
07	BEL	27	'	47	G	67	g
08	BS	28	(	48	H	68	h
09	HT	29	)	49	I	69	i
0A	LF	2A	*	4A	J	6A	j
0B	VT	2B	+	4B	K	6B	k
0C	FF	2C	,	4C	L	6C	l
0D	CR	2D	-	4D	M	6D	m
0E	SO	2E	.	4E	N	6E	n
0F	SI	2F	/	4F	O	6F	o
10	DLE	30	0	50	P	70	p
11	DC1	31	1	51	Q	71	q
12	DC2	32	2	52	R	72	r
13	DC3	33	3	53	S	73	s
14	DC4	34	4	54	T	74	t
15	NAK	35	5	55	U	75	u
16	SYN	36	6	56	V	76	v
17	ETB	37	7	57	W	77	w
18	CAN	38	8	58	X	78	x
19	EM	39	9	59	Y	79	y
1A	SUB	3A	:	5A	Z	7A	z
1B	ESC	3B	;	5B	[	7B	{
1C	FS	3C	<	5C	\	7C	
1D	GS	3D	=	5D	]	7D	}
1E	RS	3E	>	5E	^	7E	~
1F	US	3F	?	5F	_	7F	DEL

(1) 032600 000000 11 00 10 0110 0000 0000

(B) = 006000

(PC) = 003000

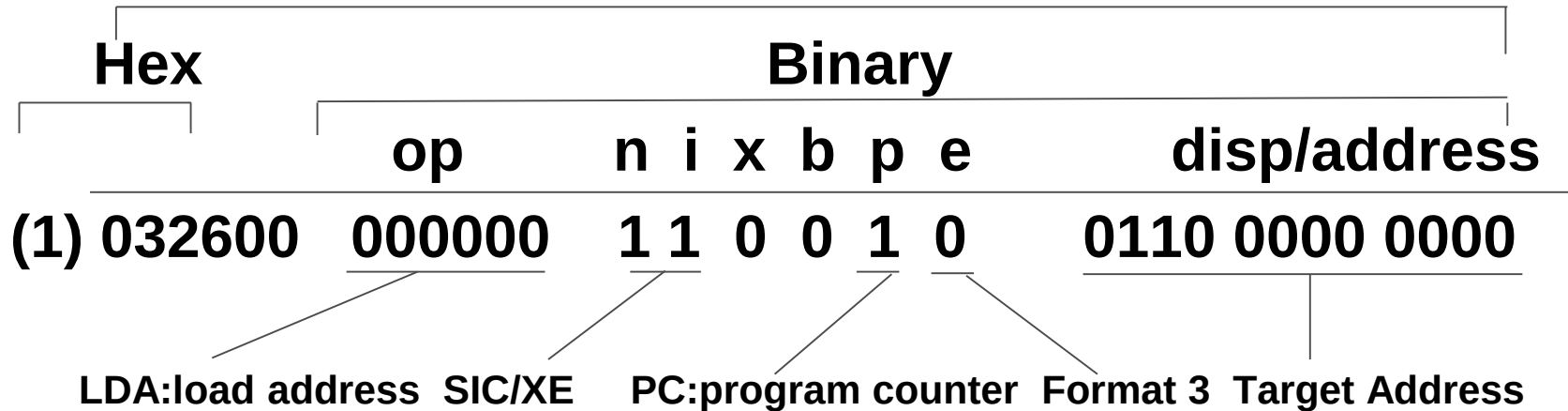
(X) = 000090

000000 110010 0110 0000 0000

Simple, PC-relative

Hex: 0 3 2 6 0 0

Machine instruction



TA(Target Address) : 003000(Program Counter) + 600(Displacement) = 3600

p11. figure(a) : 3600 103000

Value loaded into register A: 103000

(2) 03C300 000000 1 1 1 1 0 0 0011 0000 0000

(B) = 006000

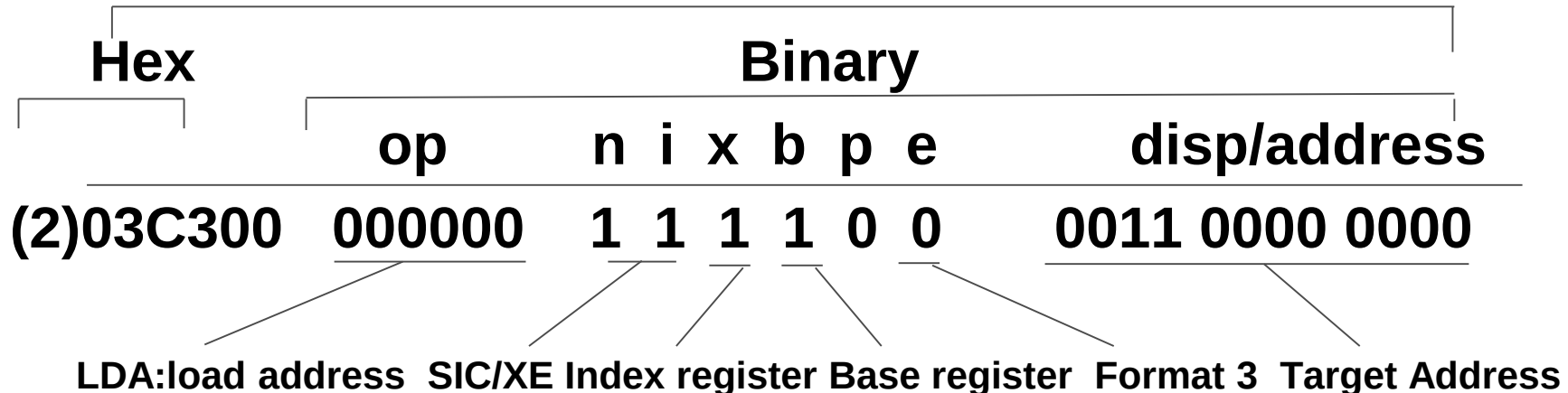
(PC) = 003000

(X) = 000090

000000 111100 0011 0000 0000
Hex: 0 3 C 3 0 0

Simple, index, base-relative

Machine instruction



Target address : 000090(Index register) + 300(Displacement) + 006000(Base register) = 6390

p11. figure(a) : 6390 00C303

Value loaded into register A: 00C303

(3)022030 000000 1 0 0 0 1 0 0000 0011 0000

(B) = 006000

(PC) = 003000

(X) = 000090

000000 100010 0000 0011 0000

indirect, program-counter relative

Hex: 0 2 2 0 3 0

Machine instruction

Hex

Binary

	op	n	i	x	b	p	e	disp/address
(3)022030	000000	1	0	0	0	1	0	0000 0011 0000

LDA:load address SIC/XE

Program counter

Format 3 Target Address

Target address : 003000(Program Counter) + 030 = 3030 (Indirect Address)

Load address: 3600

p11. figure(a) : 3600 103000

Value loaded into register A: 103000

3030 003600

(4)010030 000000 0 1 0 0 0 0 0000 0011 0000

(B) = 006000

(PC) = 003000

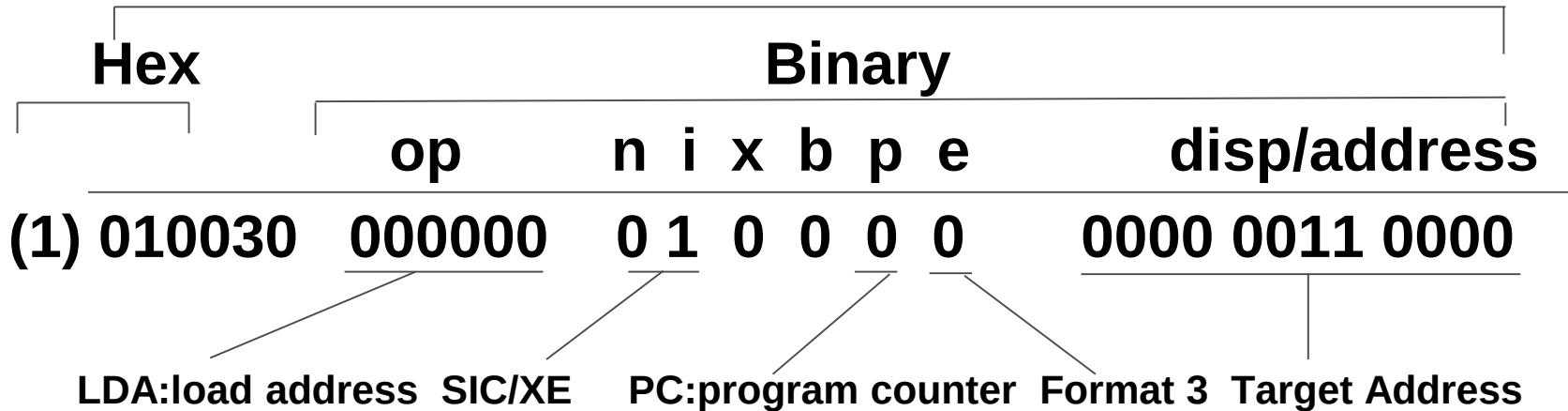
(X) = 000090

000000 010000 0000 0011 0000

immediate

Hex : 0 1 0 0 3 0

Machine instruction



TA(Target Address) : + 030 = 30

Value loaded into register A: 000030

(5)003600 000000 0 0 0 0 1 1 0110 0000 0000

(B) = 006000

(PC) = 003000

(X) = 000090

000000 000011 0110 0000 0000

Hex: 0 0 3 6 0 0

Machine instruction

SIC, direct, format 3

Hex

Binary

	op	n	i	x	b	p	e	disp/address
(5)003600	000000	0	0	0	0	1	1	0110 0000 0000

LDA:load address SIC

Target Address

Target address : (direct address) 3600

p11. figure(a) : 3600 103000

Value loaded into register A: 103000

(6)0310C303 000000 1 1 0 0 0 1 0000 1100 0011 0000 0011

(B) = 006000

(PC) = 003000

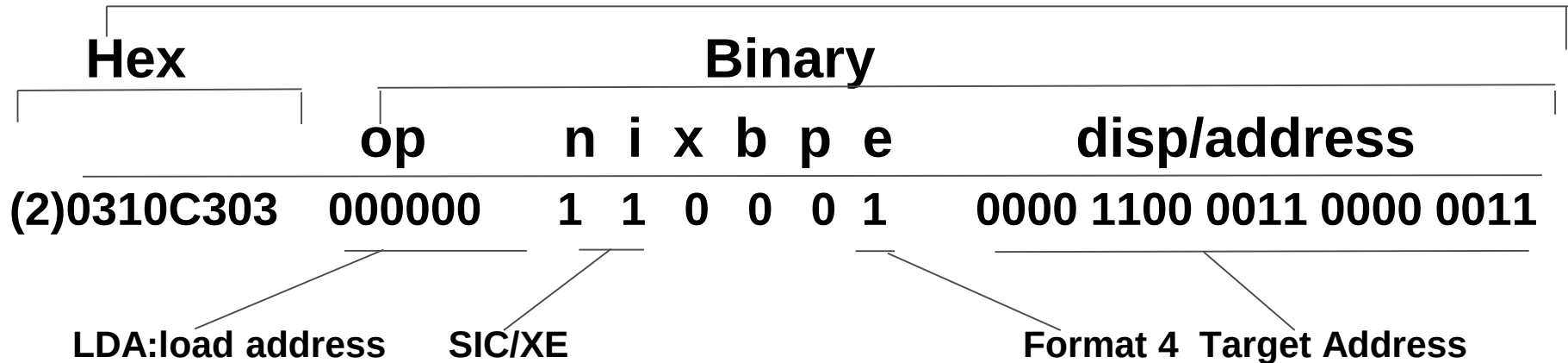
(X) = 000090

000000 110001 0000 1100 0011 0000 0011

Simple, format 4

Hex: 0 3 1 0 C 3 0 3

Machine instruction



Target address : 0C303

p11. figure(a) : C303 003030

Value loaded into register A: 003030

Project 1

입력 HEX 코드에 대해 다음과 같은 조건의 결과를 출력하는 프로그램을 작성

1. Binary code
2. Opcode
3. Flag bit
4. Address mode
5. Disp/Addr
6. TA
7. Register A value

```
YongyunCho@DESKTOP-681VVI2 ~  
$ !~  
~/a.exe  
Hex 입력 : 032600  
Binary : 000000110010011000000000  
Opcode : 000000  
nixbpe : 110010 (n=1 i=1 x=0 b=0 p=1 e=0)  
Flag bit : SIC/XE, Simple, PC-relative, Format 3  
disp/addr : 011000000000  
Target Address = 0x3600  
Register A value = 0x103000  
YongyunCho@DESKTOP-681VVI2 ~  
$
```

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지자체-대학 협력기반 지역혁신 사업비로 개발되었음.